Supplementary Information

Multiple intramolecular triggers converge to preferential G protein coupling in the CB_2R

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Supplementary Note 1 - Detailed structural description of PrefCoup$_{G\alpha i2}$ cluster 1 to 3

In the PrefCoup$_{G\alpha i2}$ cluster 1, logistic regression highlights differences in contact stability and dynamics for three conserved domains (**Figure 5B**): (1) The conserved W258^{6x48}, also known as the toggle switch, within the CWxP motif in TM6 shows an increased interaction frequency with residue L195^{5x44} in $G_{\alpha i2}$ preferential mutants. (2) Two residues in the sodium binding site (D80^{2x50} and S120^{3x39}) show an increased interaction stability in $G_{\alpha i2}$ preferential mutants. (3) Finally, we also find differences involving the E/DRY motif in the intracellular half of TM3. Two consecutive residues from this motif, R131^{3x50} and Y132^{3x51}, show a decreased interaction stability with residue V212^{5x61} in $G_{\alpha i2}$ preferential mutants.

Interestingly, in the PrefCoup$_{G\alpha i2}$ cluster 2 (**Figure 5C**), we identify alterations in contact dynamics for the same motifs as in cluster 1, but with different residues implicated: (1) The toggle switch (W258^{648}) is involved in two interactions with C288^{7x41} and N291^{7x45} that lose stability in $G_{\alpha i2}$ preferential mutants. The latter residues are also part of the sodium binding site. (2) In the sodium binding site (besides contacts with N291^{7x45}), residue S292^{7x46} is establishing contacts with S47^{1x46} which are more stable in $G_{\alpha i2}$ preferential mutants. (3) Finally, the DRY motif shows a modified interaction involving residues R131^{3x50} and T127^{3x46}. This interaction is established between stacked residues in the helical structure of TM3 and is more stable in $G_{\alpha i2}$ preferential mutants. Moreover, we have identified a densely populated region of highlighted interactions in the extracellular halves of TM7, 1, and 2, comprising residues C40^{1x39}, F91^{2x61}, F87^{2x57}, S285^{7x38}, and G44^{1x43}. Generally, the interactions within this network are less stable in Gαi2 preferential mutants, with only one interaction showing increased stability, i.e. between residues C40^{1x39} and F87^{2x57}.

In PrefCoup$_{G\alpha i2}$ cluster 3 (**Figure 5D**), the logistic regression model highlights differences in the NPxxY motif (1) and the sodium binding site (2). The NPxxY motif stands out with two highlighted inter-residual contacts that are more stable in $G_{\alpha i2}$ preferential mutants. One interaction is with residue N51^{1x50}, and the other is with residue D80^{2x50} in the sodium binding site. (3) The DRY motif also shows altered interactions in this cluster. Specifically, we observe two interactions involving residue R131^{3x50} that are more stable in $G_{\alpha i2}$ preferential mutants and are established with residues T127^{3x46} and T246^{6x36}.

Supplementary Note 2 - Molecular mechanisms of arrestin recruitment

An interesting observation of our study is that the C-terminus or intracellular loops 2 and 3 do not affect preferential coupling based on our threshold definitions. In the CB_2R, intracellular loops 2 and 3 each contain only a single serine and are both relatively short. Previous studies[1] have shown that effective arrestin recruitment requires a specific motif consisting of multiple phosphorylated residues. While such extended phosphorylation motifs may be present in GPCRs with longer intracellular loops, they are notably absent in CB_2R's intracellular loops 2 and 3. In contrast, the C-terminal region of CB_2R contains a sufficient number of serine and threonine residues that could form a phosphorylation motif capable of facilitating arrestin recruitment.

However, we find that mutating those serines or threonine residues in our study does not result in sufficient β-arrestin impairment to yield preferential G protein coupling. This could be attributed to the fact that we focus on mutations with a drastic impact, i.e. Emax < 50% of wild-type receptor response. Likely, single-point mutations are insufficient to lead to such a drastic decrease in the efficacy (Emax < 50%) of arrestin recruitment to CB_2R, as other phosphorylation sites are still present and compensate for the lack of the mutated one. This is in line with previous studies reporting that point mutations of serines and threonines affect arrestin recruitment, but do not necessarily abrogate it completely[2,3], including the barcode hypothesis, which suggests that differential phosphorylation patterns of GPCR C-terminus or intracellular loops lead to distinct modes of arrestin binding to the receptor[1,4]. The arrestin recruitment assay used in this study, based on ebBRET, does not report on the exact positioning of arrestin. Therefore, we cannot comment on whether the individual Ser/Thr mutations lead to distinct arrestin binding modes.

Ultimately, CB_2R has been shown to recruit β-arrestin independently of GRK-mediated phosphorylation[5]. In fact, this has been confirmed by us using GRK2/3/5/6 knockout cell lines (**Supplementary Figure 5**). In this context, negatively charged C-terminal aspartic acid residues have been proposed to act as phospho-mimetics, triggering arrestin activation—a mechanism observed in other receptors as well[4]. Alternatively, β-arrestin recruitment and activation may also occur through the receptor core alone[6], without requiring C-terminal phosphorylation. Taken together, it is not surprising that single Ser/Thr point mutations in the CB_2R intracellular region do not drastically affect agonist-induced β-arrestin recruitment.

Supplementary tables

Supplementary Table 1. Simulated mutants indicating the sequence position, GPCRdb notation, coupling profile classification

Position	GPCRdb notation	Coupling profile
33	1x32	Coup_Gαi2_βarr1
47	1x46	Coup_Gαi2_βarr1
49	1x48	Coup_Gαi2_βarr1
52	1x51	Coup_Gαi2_βarr1
53	1x52	Coup_Gαi2_βarr1
61	1x60	PrefCoup_Gαi2
77	2x47	PrefCoup_Gαi2
91	2x61	Coup_Gαi2_βarr1
109	3x28	PrefCoup_Gαi2
112	3x31	Coup_Gαi2_βarr1
117	3x36	PrefCoup_Gαi2
119	3x38	Coup_Gαi2_βarr1
121	3x40	Coup_Gαi2_βarr1
125	3x44	PrefCoup_Gαi2
152	4x44	Coup_Gαi2_βarr1
159	4x51	Coup_Gαi2_βarr1
165	4x57	Coup_Gαi2_βarr1
176	176	PrefCoup_Gαi2
178	178	Coup_Gαi2_βarr1
185	185	Coup_Gαi2_βarr1
199	5x48	PrefCoup_Gαi2
203	5x52	Coup_Gαi2_βarr1
205	5x54	PrefCoup_Gαi2
217	5x66	PrefCoup_Gαi2
251	6x41	Coup_Gαi2_βarr1
282	7x35	Coup_Gαi2_βarr1
285	7x38	PrefCoup_Gαi2
291	7x45	PrefCoup_Gαi2
292	7x46	PrefCoup_Gαi2
293	7x47	Coup_Gαi2_βarr1
297	7x51	PrefCoup_Gαi2
302	7x56	PrefCoup_Gαi2
313	8x57	Coup_Gαi2_βarr1
319	8x63	Coup_Gαi2_βarr1

Supplementary Table 2. N7x45A mutation in the δ opioid receptor shows differential impact on agonist function for $G_{\alpha i2}$ activation. This data has been compiled from the Supplementary Table 2 in Fenalti *et al.* Molecular control of δ-opioid receptor signalling, Nature 506, 191–196 (2014)

	ligand	Emax ± SEM	pEC50 ± SEM
WT	BW373U86	100.0 ± 1.5	8.44 ± 0.03
	DADLE	102.2 ± 1.3	8.90 ± 0.03
N7x45A	BW373U86	100.7 ± 8.7	7.99 ± 0.18
	DADLE	N/A	N/A

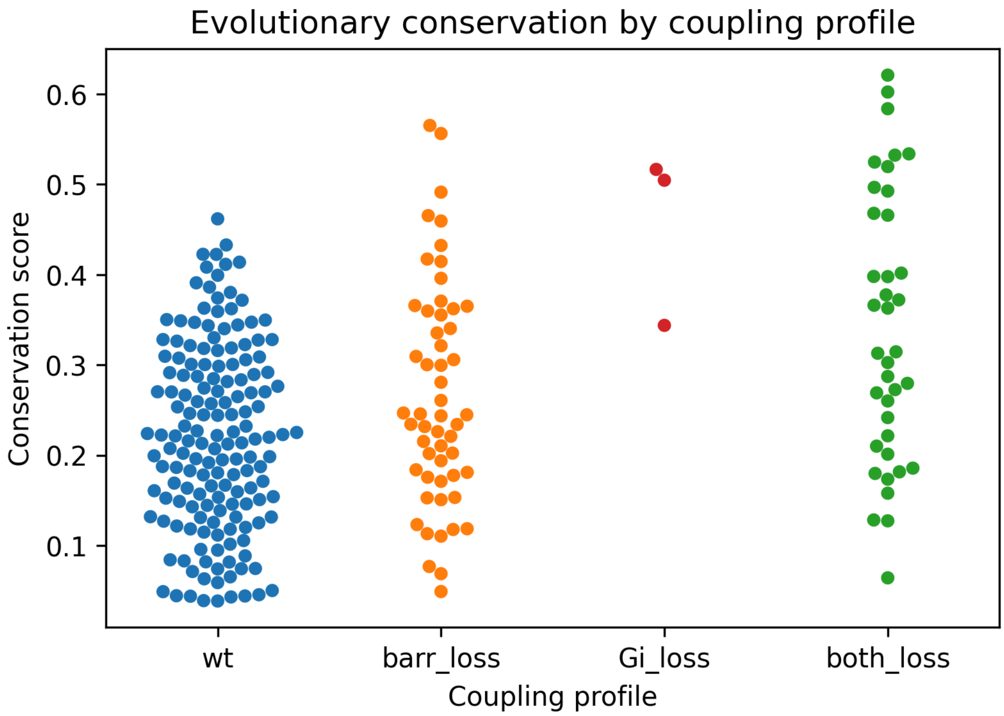
Supplementary Table 3. Reliability and reproducibility checklist for molecular dynamics simulations

Reliability and reproducibility checklist for molecular dynamics simulations ***All boxes must be marked YES by acceptance unless an N/A option is available**	**Yes**	**N/A**	**Response (Please state where this information can be found in the text)**
1. Convergence of simulations and analysis			
1a. Is an evaluation presented in the text to show that the property being measured has equilibrated in the simulations (*e.g.* time-course analysis)?	☒		Simulation time for ensuring sufficient equilibration and sampling was used according to state-of-the-art protocols (Rodríguez-Espigares I, Torrens-Fontanals M, Tiemann JKS, et al. GPCRmd uncovers the dynamics of the 3D-GPCRome. Nat Methods. 2020 Aug;17(8):777-787) Statement is found in method section/simulation protocol.
1b. Then, is it described in the text how simulations are split into equilibration and production runs and how much data were analyzed from production runs?	☒		Statement is found in method section/simulation protocol.
1c. Are there at least 3 simulations per simulation condition with statistical analysis?	☒		Five replicates are used. Statistics are found in Figure 3d, 3e Supplementary Figure 13 Supplementary Figure 14

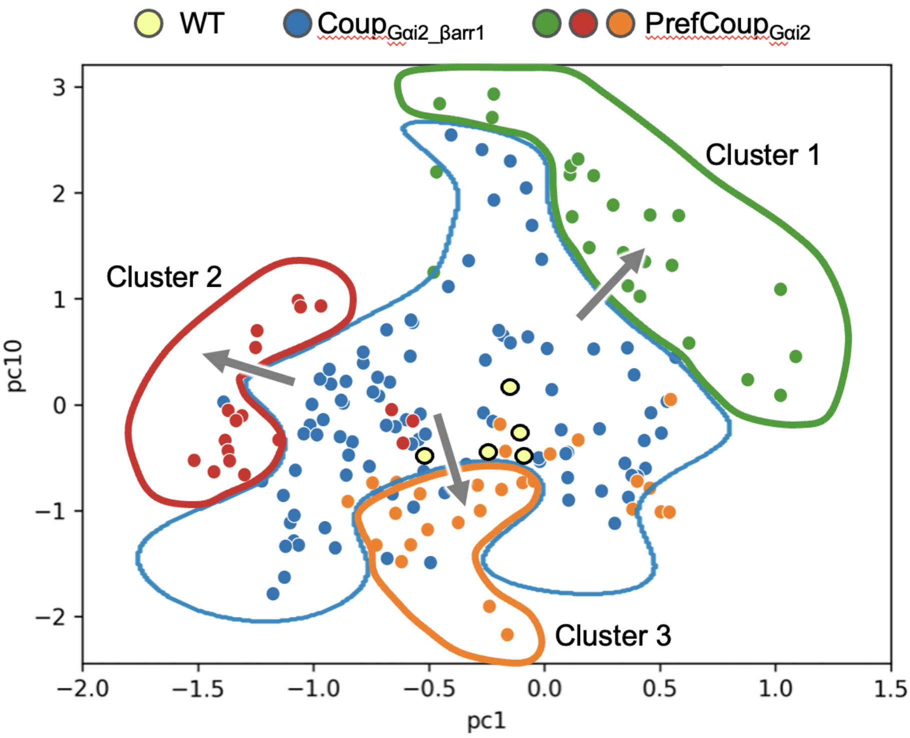
1d. Is evidence provided in the text that the simulation results presented are independent of initial configuration?	☒		The simulation time and replicates are sufficient to allow the system to comprehensively explore the conformational space around the experimental starting structure which minimizes the dependence on the initial configuration. This is stated in the method section.
2. Connection to experiments			
2a. Are calculations provided that can connect to experiments (*e.g.* loss or gain in function from mutagenesis, binding assays, NMR chemical shifts, J-couplings, SAXS curves, interaction distances or FRET distances, structure factors, diffusion coefficients, bulk modulus and other mechanical properties, *etc.*)?	☒		Intramolecular network analysis from MD data are linked to experimental mutational data and signaling bias (Figure 3).
3. Method choice			
3a. Is it described in the text what force field and water model are used and why?	☒		In method section/simulation protocol
3b. Do simulations contain membranes, membrane proteins, intrinsically disordered proteins, glycans, nucleic acids, polymers, or cryptic ligand binding?	☒	☐	In method section/simulation protocol
If 3b is **YES**, are enhanced sampling methods used?	☐	☒	
If enhanced sampling methods are used, are the convergence criteria clearly stated?	☐		n/a
If 3b is **YES**, is it explained in the text why or why not enhanced sampling methods are used?	☒		In methods section/simulation protocol
4. Code and reproducibility			

4a. Is a table provided describing the system setup, such as simulation box dimensions, total number of atoms, total number of water molecules, salt concentration, lipid composition (number of molecules and type)?	☒		The paper describes a large database of simulations (WT and 35 different mutant systems). System composition (lipids, ions, waters, number of molecules), starting structure, and simulation protocols are provided here for each individual system: https://gpcrmd.org/dynadb/publications/1540/
4b. Is it described in the text what simulation and analysis software and which versions are used?	☒		Details provided in the methods section.
4c. Are initial coordinate and simulation input files and a coordinate file of the final output provided as supplementary files or in a public repository?	☒		All these files are uploaded to the public repository GPCRmd (www.gpcrmd.org): https://gpcrmd.org/dynadb/publications/1540/
4d. Is there custom code or custom force field parameters?	☒	☐	
If **YES**, are they provided as supplementary profiles or in a public repository?	☒		The python code for data processing is available at GitHub: https://github.com/GPCRmd/prefcoup_cb2r

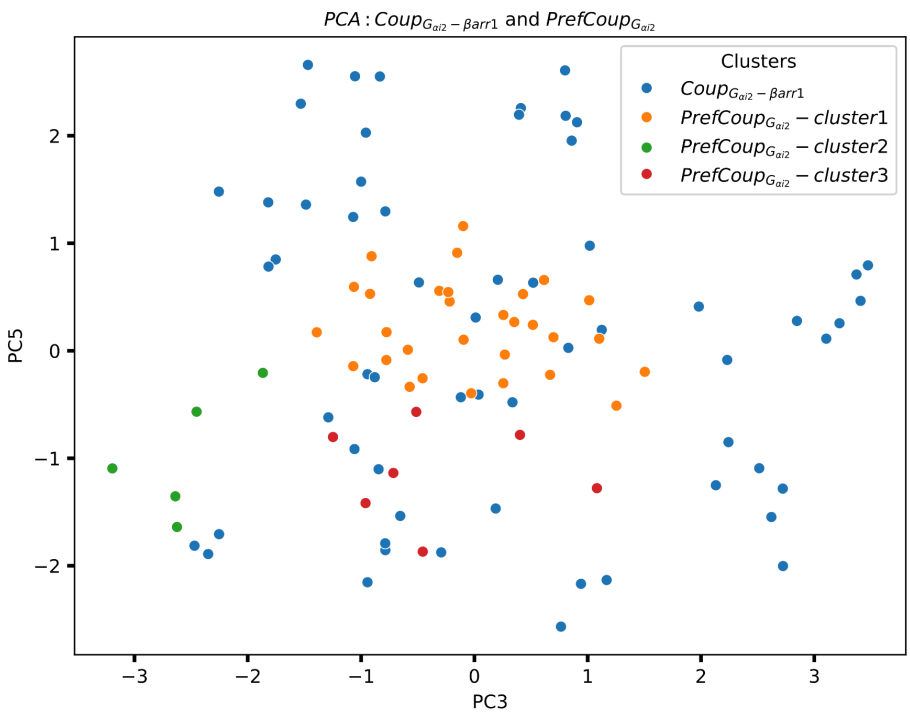
Supplementary figures



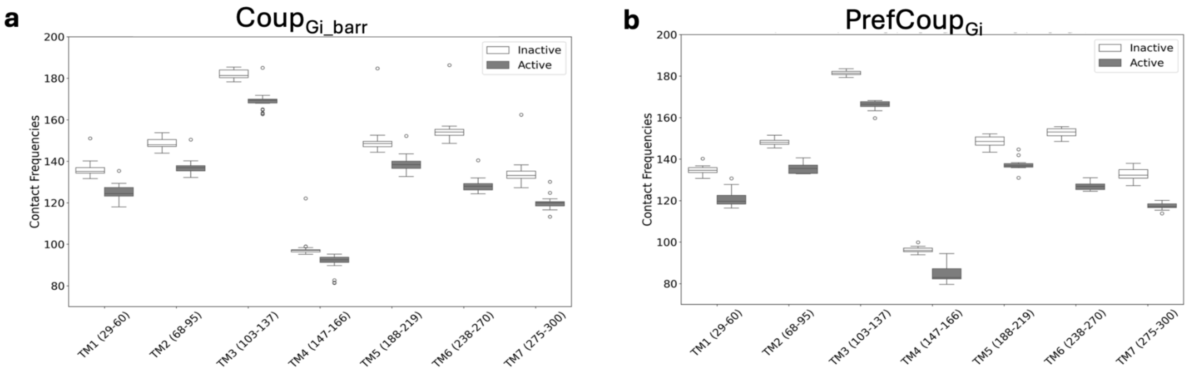
Supplementary Figure 1. Evolutionary conservation by coupling profile. This swarm plot shows the distribution of conservation scores in each one of the coupling profiles. The conservation score is computed as the Jensen Shannon Divergence of a class A GPCR multiple sequence alignment.



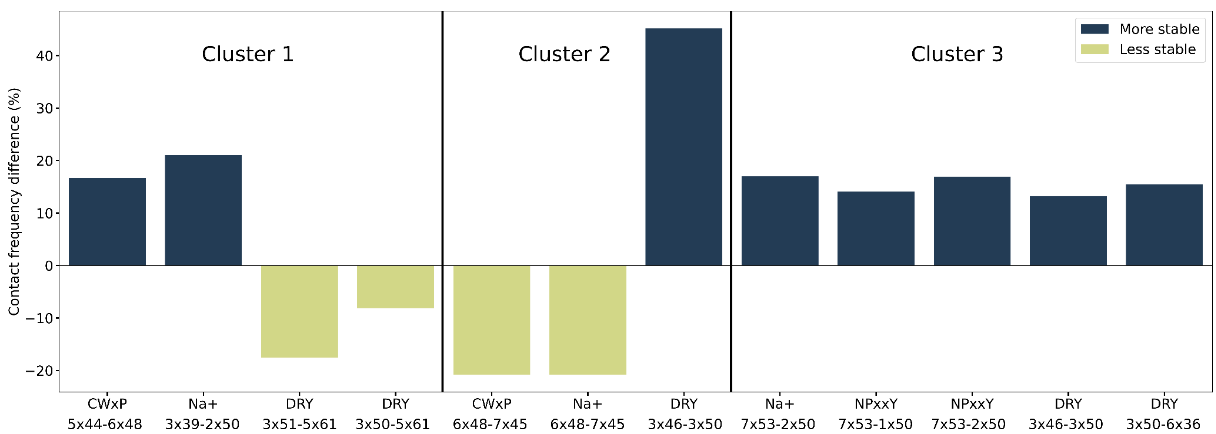
Supplementary Figure 2:
K-means clustering showing cluster 1 (green), 2 (red) and 3 (orange) for PrefCoup$_{Gai2}$ mutants and the Coup$_{G\alpha i2_barr}$ mutant group (blue).



Supplementary Figure 3. Active CB$_2$R systems in complex with the G protein - a scatterplot of the two best PCs based on the highest SVC score. The PCA is not able to separate the different mutant groups. A clustering using k-means shows that cluster 1 (orange), 2 (green), and 3 (red) of PrefCoup$_{G\alpha i2}$ group are not well separated from the Coup$_{G\alpha i2_barr}$ (blue) group. This is likely due to the G protein penetration into the receptor core which significantly reduces intramolecular contacts in the transmembrane receptor domain (see **Supplementary Figure 4**).

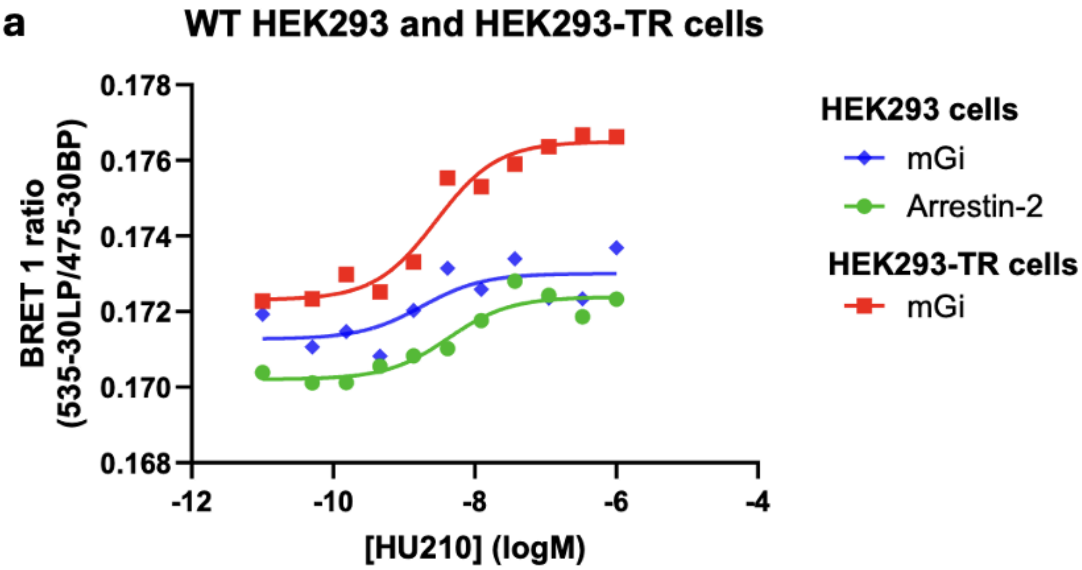


Supplementary Figure 4. Contact profile for the transmembrane receptor domain (TM1 to TM7) comparing inactive versus the active CB_2R systems for the (**a**) PreCoup$_{G\alpha i2}$ and (**b**) Coup$_{G\alpha i2_barr}$ mutant group. For this, we summed the contact frequencies of residues within each TM segment (i.e., TM1 to TM7) for each mutant. We then generated boxplots for TM1 to TM7, which highlight the contact distribution across mutants for inactive and active CB_2R systems. Each TM helix is followed by the corresponding residues in brackets, indicating the specific residues for which contact frequencies have been aggregated. We find that there is a general contact loss in the transmembrane domain (specifically TM6) for active receptor mutants which can be attributed to the G protein penetration into the receptor core. This contact loss overall reduces the allosteric communication in the CB_2R core compared to inactive receptor systems. This likely explains the unsuccessful performance of the PCA model in capturing the mutant-induced impact on the allosteric communication network in the active CB_2R (see **Supplementary Figure 3**).

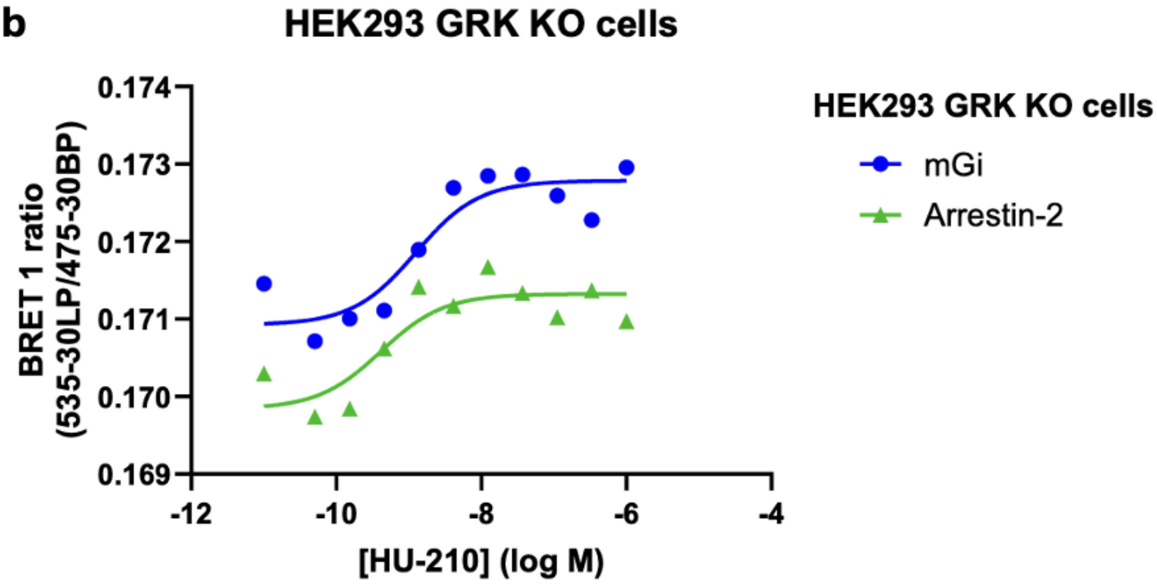


Supplementary Figure 5. Delta contact frequency (ΔCF) of contacts selected by the regression model for discriminating PrefCoup$_{G\alpha i2}$ from Coup$_{G\alpha i2_\beta arr1}$ mutants. For this, we have computed the mean CF for selected contacts of each PrefCoup$_{G\alpha i2}$mutants cluster and subtracted the mean CF of Coup$_{G\alpha i2_\beta arr1}$ mutants (i.e., ΔCF = CF_PrefCoup$_{G\alpha i2}$ - CF_Coup$_{G\alpha i2_\beta arr1}$). If the contact is more stable in the PrefCoup$_{G\alpha i2}$ than in the Coup$_{G\alpha i2_\beta arr1}$, the ΔCF is positive and vice versa. Each bar is color-coded based on the predictions from a logistic regression model, with navy blue representing more stable contacts and yellow indicating less stable contacts for PrefCoup$_{G\alpha i2}$ compared to CoupG$_{\alpha i2_\beta arr1}$ mutants. The ΔCF values align well with the predicted coefficients. It is important to note that contacts that are captured by the logistic regression might not contribute alone to the model but are involved in a network of correlated contacts.

a



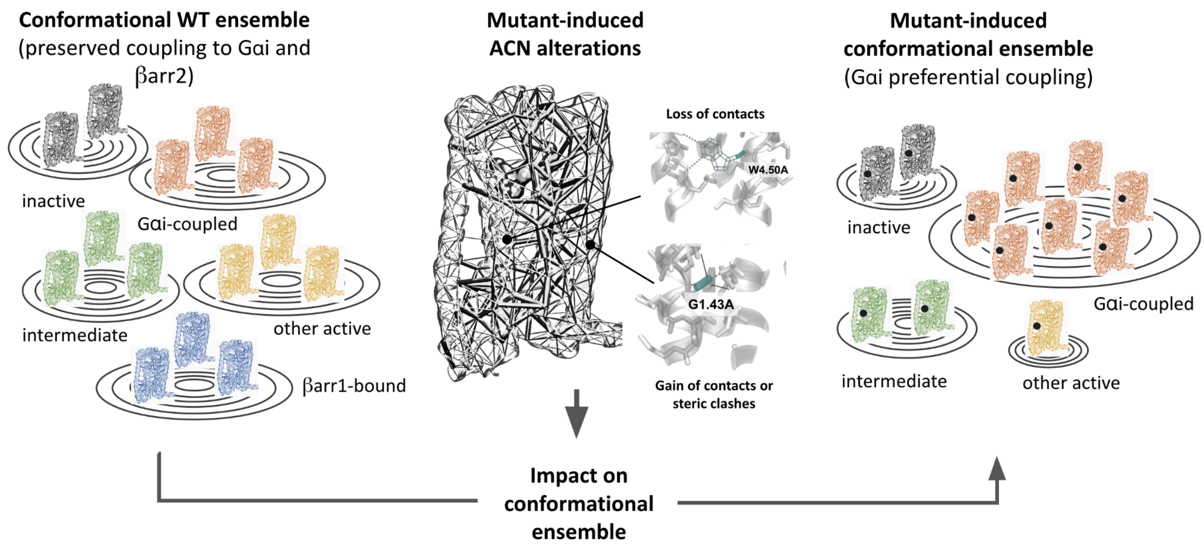
b



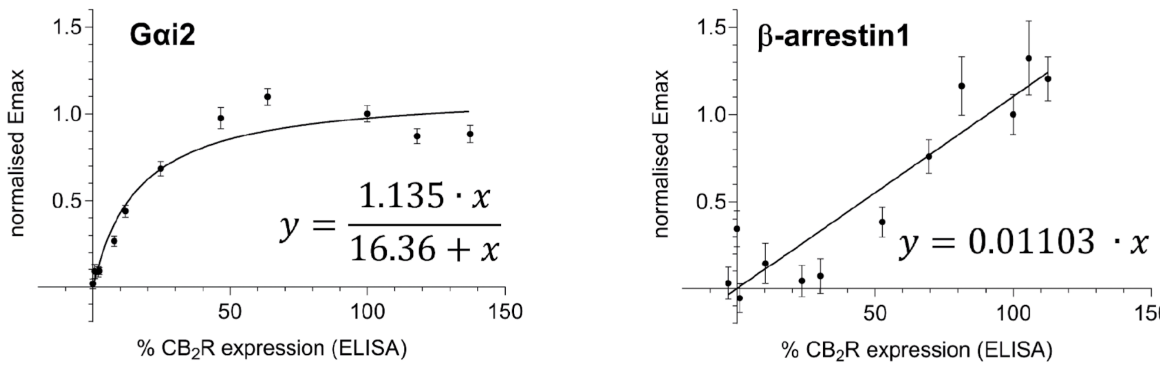
Supplementary Figure 6. BRET-based CB_2R mini Gi and β-arrestin2 recruitment assays. **(a)** CB_2R mini G_i (mG_i) and β-arrestin2 concentration response curves to the CB_2R agonist HU210. Data were obtained in HEK293 cells and in T-REx^{TM}-293 cells stably expressing either mG_i or β-arrestin2 and transiently transfected (48h) with the human CB_2R-nLuc construct. Data shown are from a single experiment. (**b**) CB_2R mG_i and β-arrestin2 concentration response curves to the CB_2R agonist HU210. Data were obtained in HEK293 GRK KO cells stably expressing either mG_i or β-arrestin2 and transiently transfected (48h) with the human CB_2R-nLuc construct. Data shown are from a single experiment. **Materials:** The T-REx^{TM}-293 cell line was obtained from Invitrogen (CA, U.S.A). HEK293 GRK KO cells were a generous gift by Carsten Hoffmann lab. All cell culture reagents, including Phosphate Buffered Saline (PBS) and Fetal Calf Serum (FCS) were purchased from Sigma Aldrich (Gillingham, UK), except for Blasticidin and Zeocin^{TM} which was obtained from Gibco™ (MA, U.S.A). Polyethylenimine (PEI) (25kDa) was obtained from Polysciences Inc (PA, U.S.A), and culture plates from Greiner Bio-One (code 655098 Kremsmünster, Austria). **Mini-G protein and β-arrestin recruitment assays.** HEK293, T-REx™-293 and HEK293 GRK KO cells stably expressing fluorescently labeled mini G_i (mG_i) protein and β-arrestin2 were used to assess effector recruitment. Cultured cells were harvested upon reaching 70% confluency, resuspended at a density of 0.5 million/mL and transfected with 0.5 µg/mL of pcDNA4_CB2R-nLuc plasmid using a PEI:DNA ratio of 4:1. Cells were plated at a seeding density of 50,000 cells per well in Poly-D-Lysine coated 96-well plates. The cells were grown for 48h until they reached confluency and then stimulated with 1 µg/mL tetracycline for a further 48h to induce expression. Cell culture media was then aspirated from the wells and the cells were washed with assay buffer, 100 µL/well Hanks′ Balanced Salt solution (HBSS) containing 0.5% BSA and 5 mM HEPES. Following the wash, assay buffer containing 10 µM furimazine, 90 µL/well was added to the wells. The plate was then incubated at 37ºC for 15 minutes to allow the nLuc substrate furimazine to fully equilibrate, then transferred to the Pherastar FSX preset to 37ºC. Three BRET cycles of 1-minute intervals were performed to assess basal nanoBRET levels, following which 10 µL of each compound, diluted in assay buffer was added to the assay plate, which was read at 1-minute intervals for 30 minutes. HU210 was serial diluted in DMSO in polypropylene plates (100x final concentration) then transferred to a second 96-well dilution containing assay buffer (10x final concentration and 10% DMSO). Finally, compounds were transferred to the assay plates. 1% DMSO in the assay buffer served as the vehicle control. Responses were monitored for up to 90 min in the case of mG_i and 60 min in the case of β-arrestin2. **Concentration response curves.** mG_i and β-arrestin2 concentration response data were fitted to sigmoidal (variable slope) curves using a “four parameter logistic equation”:

$$Y = Bottom + \frac{(Top - Bottom)}{1+10^{(logEC50-X)*Hillslope}} \qquad \text{Eq.1}$$

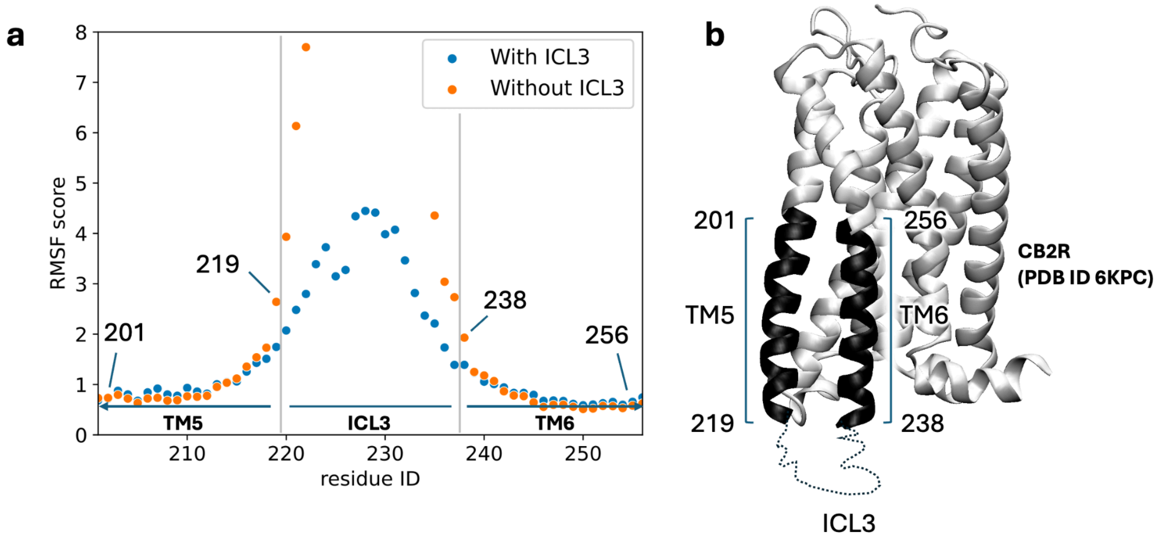
Where, Bottom and Top are the lower and upper plateaus of the agonist concentration response curves. $LogEC_{50}$ is the concentration of agonist that gives a half-maximal effect, and the Hillslope is the unitless slope factor or Hillslope.



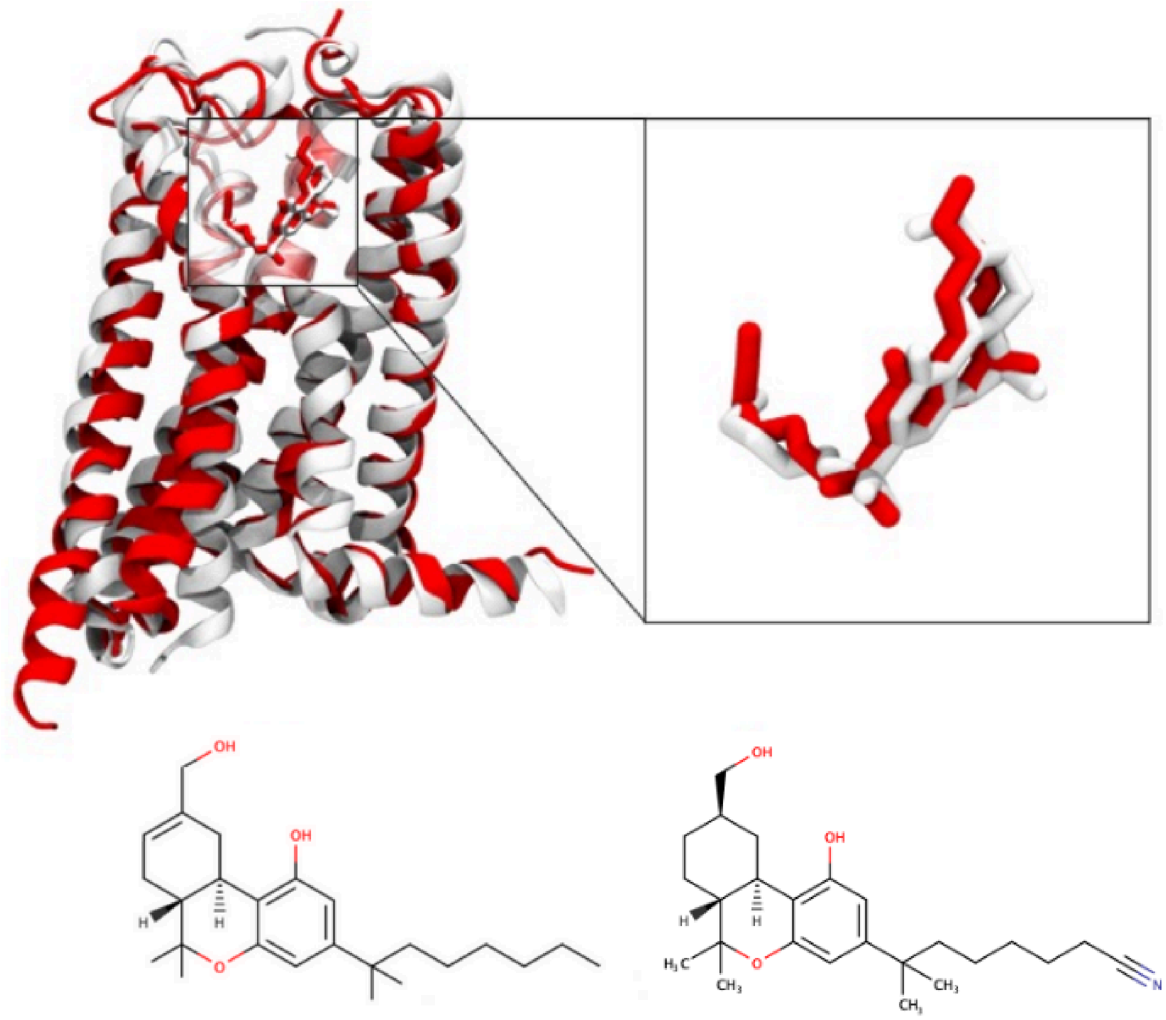
Supplementary Figure 7. Schematic model of mutant-induced alterations of allosteric communication networks (ACN) and their impact on the conformational ensemble comprising inactive, intermediate, and various active states.



Supplementary Figure 8. Correlation of CB_2R expression levels with measured Emax values, used to calculate the correction factors in Supplementary Data 1.

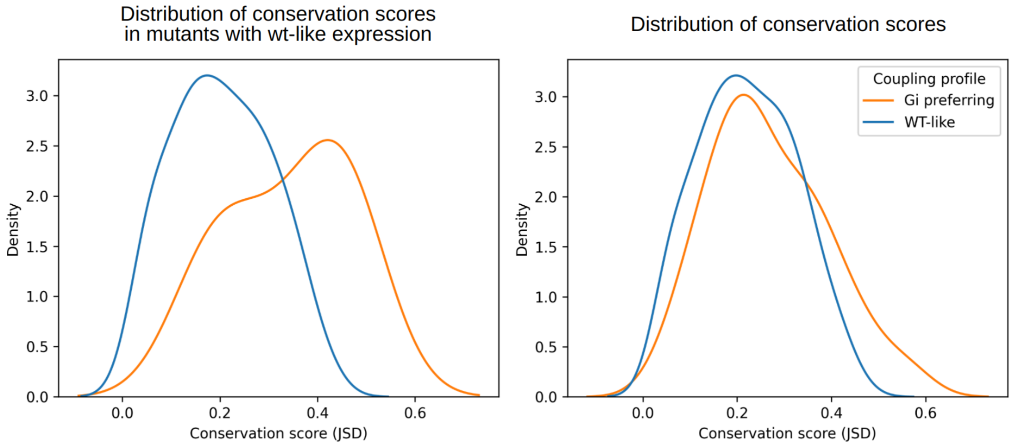


Supplementary Figure 9. (**a**) Root Mean Square Fluctuation (RMSF) of the ICL3 loop region for the inactive CB_2R with the loop modelled (with loop, blue) versus the loop capped (without loop, orange). (**b**) Crystallised structure of the CB_2R receptor (PDB ID 6KPC) highlighting the experimentally solved TM5 and TM6 and their connection via the unsolved ICL3 loop region.

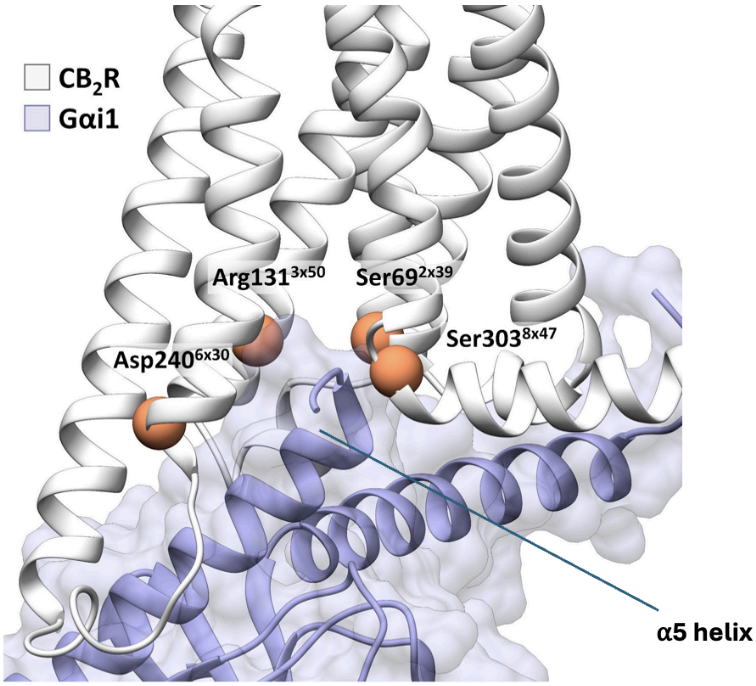


HU-210 AM12033

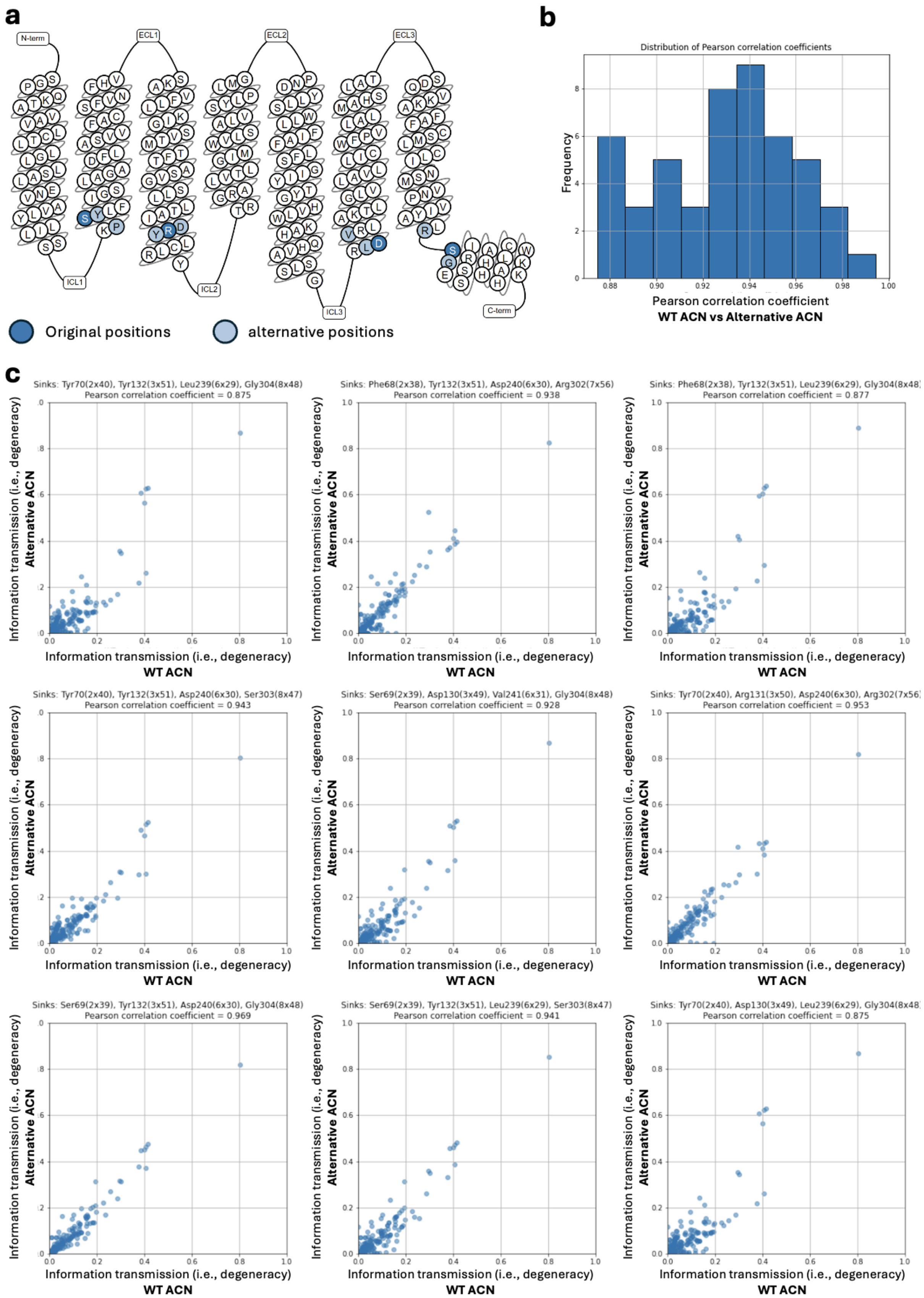
Supplementary Figure 10. Comparison of CB2 Receptor (CB_2R) Conformations and Ligand Docking Models. The white ribbon diagram represents the inactive CB_2R structure modelled from PDB entry 5ZTY[7], while the red ribbon diagram overlaid represents the inactive CB_2R conformation from PDB entry 6KPC[8]. Highlighted within the receptor is the compound HU-210 docked into the inactive structure, positioned to mimic the binding pose of the red molecule, AM12033, which is docked in the active structure. Below are the 2D chemical structures of HU-210 and AM12033, illustrating their molecular resemblance.



Supplementary Figure 11. Evolutionary conservation in mutants that don't affect expression. Comparison of the distribution of conservation on coupling profiles of interest between residues that do not impact expression against all the residues. Conservation scores are computed using the Jensen Shannon divergence on an MSA of class A GPCRs.

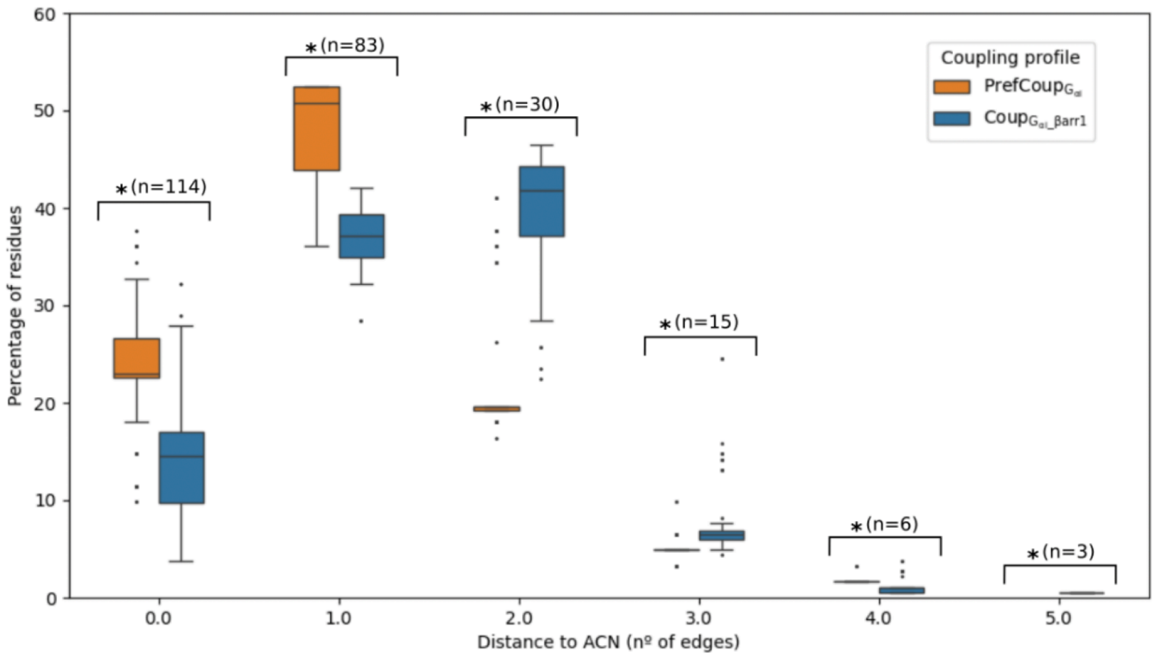


Supplementary Figure 12. Structural depiction of the CB_2R $G_{\alpha i}$ complex (PDB ID: 6KPF[8]). Residues at the transmembrane ends of CB_2R that are in close proximity to α5 helix of the $G_{\alpha i}$ protein (CA atoms are shown in orange spheres) are used as a sink for the computation of the shortest pathways (see method section in main text): Ser69 (2x39), Arg131 (3x50) and Ser303 (8x47) are polar residues located in the interface between the α5 helix of Gαi and the intracellular domain of the active CB_2R (PDB ID: 6KPF[8]). Of note, their interactions are present across the different CB_2R-$G_{\alpha i}$ structures (PDB IDs: 6KPF[8], 6PT0[9], 8GUS[10], 8GUT[10], 8GUQ[10] and 8GU[10]), confirming their relevance for G protein complex formation. Ultimately, Asp240 (6x30) is crucial for receptor activation as its interaction with Arg131 (3x50) forms the “ionic lock”, which maintains the receptor in the inactive state. Therefore, information must be transmitted to this residue from the ligand binding site during receptor activation to disrupt the ionic lock.

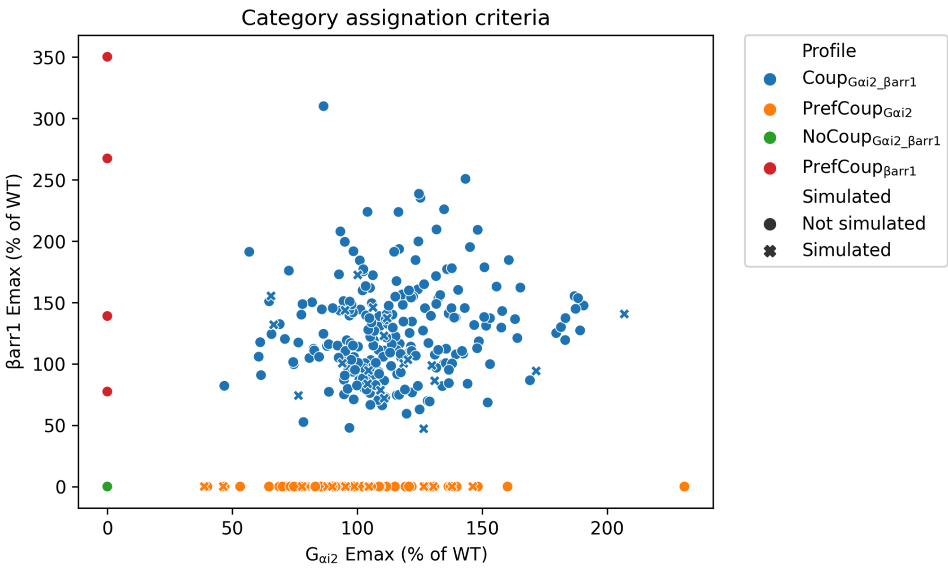


Supplementary Figure 13. Evaluation of the robustness of ACN computation from MD simulation. (**a**) Snake plot showing the originally selected (dark blue) and alternative (light blue) sinks for the calculation of allosteric networks. To evaluate robustness, we calculated

50 random "alternative" WT networks, where at least one of the sinks was exchanged for the residue directly above or below the ones originally selected for the analysis. (**b**). Distribution of Pearson correlation coefficients for the original WT ACN against 50 alternative networks. We evaluated the similarity between ACNs based on the information transmission for each edge in the network (i.e., their degeneracies) and calculated the Pearson correlation coefficients. The lowest Pearson correlation coefficient between the original WT network and an alternative network is 0.875, demonstrating the robustness of the applied approach. (**c**). Scatter plots of information transmission (computed as degeneracy) for the edges in the original WT network versus nine representatives from the 50 alternative networks.



Supplementary Figure 14. Distance of $PrefCoup_{G\alpha i2}$ and $Coup_{G\alpha i2\_\beta arr1}$ mutants to $LigACN_{top}$. The size and definition of the $LigACN_{top}$ depends on the cutoff for information transmission (i.e., degeneracy). Here we plot the mutant distance for different $LigACN_{top}$ definitions (degeneracy range: 0 to 0.02). The distance is represented as the number of edges from a mutation position to the $LigACN_{top}$ in the x-axis. The percentage of mutants at a given distance is provided in the y-axis. Statistical analysis was performed for each pair of $PrefCoup_{G\alpha i2}$ and $Coup_{G\alpha i2\_\beta arr1}$ mutants at each distance using Student's t-test (*$p < 0.001$).* The sample size (n) for each pair is given in the plot.



Supplementary Figure 15. Coupling profile categorization criteria. Scatterplot showing the Emax distribution of mutants for $G_{\alpha i2}$ and βarr_1 coupling. Mutants are coloured based on their coupling profile as described in Figure. A large fraction of mutants with preserved coupling ($Coup_{G\alpha i\_\beta arr1}$) and $G_{\alpha i2}$ preferential mutants ($PrefCoup_{G\alpha i}$) have been simulated. Those mutants are represented with a dot. Mutants highlighted with an 'x' are not simulated as they do not fulfill the following conditions: (1) cell surface expression similar to the WT level and (2) located in the receptor region with a solved 3D structure.

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